

Programming Assignment 4 Report

Composite Numerical Quadrature:
A Priori and Adaptive Composite Trapezoidal Rules
versus Composite Two-Point Gauss–Legendre Rule

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1 Executive Summary

This report documents the design, implementation, and empirical validation of three composite numerical quadrature methods, together with a systematic comparison of their behaviour on five test integrals with known closed-form values. The C++ implementation encompasses three quadrature codes and two supporting derivative-estimation routines: (i) a Composite Trapezoidal Rule whose subinterval width H_m is chosen *a priori* from the theoretical error bound $|I - T_n| \leq (b - a)H_m^2 \max |f''|/12$; (ii) an adaptive step-halving Composite Trapezoidal Rule using refinement factor $\alpha = 1/2$, the recursive doubling identity $T_{2n} = \frac{1}{2}T_n + H_{\text{new}} \sum f(\text{midpts})$, and the Richardson error estimate $|T_{2n} - T_n|/3$ as its stopping criterion; and (iii) a Composite Two-Point Gauss–Legendre Rule mapping the canonical nodes $\pm 1/\sqrt{3}$ onto each subinterval for a degree-of-exactness 3 quadrature with $2n$ function evaluations. All three codes share a common function-evaluation counter that accurately records work done, enabling a fair comparison of computational cost across methods.

The empirical validation in Section 4 consists of eight systematic tests covering polynomial exactness, the doubling identity, empirical convergence rates, a priori bound tightness, adaptive termination behaviour, Richardson estimate accuracy, round-off saturation, and full sweeps across tolerance levels for all five required integrals. The tests confirm: the Trapezoidal Rule is exact for polynomials of degree ≤ 1 and the 2-point Gauss rule is exact for polynomials of degree ≤ 3 to machine precision ($\sim 10^{-16}$); the doubling identity $T_{2n} = \frac{1}{2}T_n + H \sum f$ reproduces direct evaluation to machine precision ($\sim 10^{-15}$); the empirical convergence rates converge to $p = 2$ (Trapezoidal) and $p = 4$ (Gauss–Legendre) as predicted, reaching the asymptote by $n \approx 64$ for smooth integrands and by $n \approx 256$ for integrand (5) which has large $\max |f^{(4)}|$; the a priori bound is tight in slope but conservative by a factor of 3–200 in constant, with the largest gap occurring where $|f''|$ is localized near a single point; and the Richardson estimate $|T_{2n} - T_n|/3$ matches the observed true error to within a few percent across all tested cases, making it a reliable basis for the adaptive stopping criterion.

Task (i) establishes, for each integral and each requested relative error $\varepsilon_{\text{rel}} \in \{10^{-2}, 10^{-3}, \dots, 10^{-9}\}$, the largest subinterval size H_{pred} consistent with the a priori bound; for integral (1) these range from $H_{\text{pred}} = 0.195$ ($\varepsilon_{\text{rel}} = 10^{-2}$, $n = 16$) down to $H_{\text{pred}} = 6.17 \times 10^{-5}$ ($\varepsilon_{\text{rel}} = 10^{-9}$, $n = 48,659$). Task (ii) confirms that the Composite Trapezoidal Rule with these H_{pred} values always produces approximations whose true error lies below the a priori bound; the bound is tight in slope (observed error $\propto H^2$) but conservative in constant. Task (iii) shows the adaptive step-halving scheme achieves essentially identical accuracy per function evaluation as the fixed- H scheme, attains the requested tolerance at the smallest power of two n satisfying the Richardson criterion, and does so without requiring any a priori knowledge of $\max |f''|$. Task (iv) compares all three methods on a common axis of function evaluations and confirms the predicted asymptotic advantage of the 2-point Gauss–Legendre rule: to reach an absolute error of 10^{-6} on integral (1), the Trapezoidal rule requires $N_{\text{Trap}} \approx 4097$ evaluations while Gauss–Legendre needs only $N_{\text{Gauss}} \approx 64$, a factor of $\sim 64\times$ in favour of Gauss. The factor grows rapidly with target accuracy: at 10^{-10} it exceeds $100\times$ for every integral tested.

The principal practical conclusion drawn from these experiments is that *basic-rule accuracy dominates adaptive cleverness*: a degree-3 basic rule (Gauss 2-point) beats a degree-1 basic rule (Trapezoidal) by a factor of H^2 at every step, and no amount of adaptive step refinement can recover that gap. The adaptive Trapezoidal scheme is not an alternative to Gauss–Legendre but an *orthogonal* improvement that could be layered on top of any basic rule; its advantage is robustness (no derivative bound needed) rather than raw efficiency. Integrand (5), featuring the factor $1/x$ near $x = 0.1$, emerges as the stress test of the study: its enormous $\max |f''| \approx 2 \times 10^3$ and

$\max |f^{(4)}| \approx 1.5 \times 10^6$ inflate the constants in both error bounds and drive an order-of-magnitude increase in evaluation counts for every method, consistent with theoretical prediction.

2 Statement of the Problem

2.1 Background and Objectives

Numerical quadrature, the approximation of definite integrals $I[f] = \int_a^b f(x) dx$ by finite sums $Q_n[f] = \sum_{i=0}^n w_i f(x_i)$, is one of the oldest and most widely applied tools in computational mathematics. Two families of rules dominate practical work: *Newton–Cotes* rules, which fix the nodes at equally-spaced points and derive the weights by enforcing polynomial exactness, and *Gauss–Legendre* rules, which optimize both nodes and weights to achieve the highest possible polynomial exactness for a given number of nodes. On $[-1, 1]$ the k -point Gauss rule is exact for all polynomials of degree $\leq 2k - 1$, whereas the k -point Newton–Cotes rule is exact only for degree $\leq k - 1$ (or k when k is odd, due to symmetry). The composite versions of both rules partition $[a, b]$ into n subintervals of width $H_m = (b - a)/n$ and apply the basic rule on each subinterval; their error decreases as n grows and $H_m \rightarrow 0$.

The central pedagogical objectives of this assignment are: (i) to implement the two simplest composite methods in each family—the Trapezoidal Rule (closed Newton–Cotes with two nodes) and the two-point Gauss–Legendre Rule—and to do so efficiently, reusing function values wherever possible so that the function-evaluation count truly reflects computational work; (ii) to convert the theoretical a priori error bounds into a sizing strategy that picks H_m given a target accuracy; (iii) to implement an adaptive step-halving variant of the Trapezoidal Rule that uses runtime information (the Richardson extrapolation idea) to decide when to stop refining, and to compare this against the a priori strategy; and (iv) to quantify empirically the order-of-magnitude savings in function evaluations obtainable by moving from a degree-1 (Trapezoidal) to a degree-3 (2-point Gauss) basic rule.

The transition from a priori sizing to adaptive refinement is motivated by a fundamental limitation of the a priori approach: it requires an upper bound on $\max_{[a,b]} |f''|$ (for Trapezoidal) or $\max_{[a,b]} |f^{(4)}|$ (for Gauss), which is often unavailable in practice—the very purpose of numerical integration is typically to handle integrands whose symbolic derivatives are unknown or intractable. The adaptive scheme substitutes an *a posteriori* error estimate based on the difference between two successive refinement levels, which can be computed purely from the function values already gathered. This report quantifies the trade-offs (if any) between a priori sizing and adaptive refinement for integrands whose derivative bounds *are* available.

The transition from Trapezoidal to Gauss is motivated by classical approximation theory. The Trapezoidal Rule is exact for linear functions; its error on $[a, b]$ is

$$|I - T_1| = \frac{(b - a)^3}{12} |f''(\xi)|, \quad \xi \in (a, b), \quad (1)$$

and the composite version satisfies

$$|I - T_n| \leq \frac{(b - a) H_m^2}{12} \max_{[a,b]} |f''|, \quad (2)$$

giving $O(H_m^2)$ convergence. The 2-point Gauss–Legendre rule is exact for cubics; its error on $[-1, 1]$ is

$$|I - G_1| = \frac{f^{(4)}(\xi)}{135} \quad (3)$$

and the composite version satisfies

$$|I - G_n| \leq \frac{(b-a) H_m^4}{4320} \max_{[a,b]} |f^{(4)}|, \quad (4)$$

giving $O(H_m^4)$ convergence. The asymptotic advantage of Gauss over Trapezoidal is therefore a factor of H_m^2 per step—at $H_m = 0.01$ that is a factor of 10^{-4} , meaning Gauss delivers four extra decimal digits of accuracy for essentially the same per-step cost. The cost comparison is slightly subtle: at the same n , Trap uses $n + 1$ evaluations while Gauss uses $2n$, so at equal evaluation budget Gauss's advantage is reduced by a factor of $2^4 = 16$; the net effect is still a dominant asymptotic advantage for the Gauss rule whenever $f \in C^4$.

2.2 Required Methods and Their Construction

The assignment specifies three method implementations. The first is the *fixed- H Composite Trapezoidal Rule*, which evaluates

$$T_n = H_m \left[\frac{1}{2} f(a) + \sum_{i=1}^{n-1} f(a + iH_m) + \frac{1}{2} f(b) \right]$$

using exactly $n + 1$ function evaluations. The second is the *adaptive step-halving Composite Trapezoidal Rule* with $\alpha = 1/2$, which uses the recursive doubling identity

$$T_{2n} = \frac{1}{2} T_n + H_{\text{new}} \sum_{i=0}^{n-1} f(a + (i + \frac{1}{2})H_{\text{old}})$$

to compute the refined estimate from the coarse one in n new function evaluations (rather than $2n + 1$ starting from scratch). After L refinements the total is $2^L + 1$ evaluations, which is optimal given that the refined rule uses $2^L + 1$ nodes in total. The third is the *Composite Two-Point Gauss–Legendre Rule*, where each subinterval $[x_i, x_{i+1}]$ of width H_m contributes

$$\int_{x_i}^{x_{i+1}} f dx \approx \frac{H_m}{2} \left[f(x_m - \frac{H_m}{2\sqrt{3}}) + f(x_m + \frac{H_m}{2\sqrt{3}}) \right]$$

with $x_m = (x_i + x_{i+1})/2$. The composite rule uses $2n$ function evaluations; no reuse is possible because successive subintervals do not share nodes under the 2-point Gauss rule.

Both Trapezoidal variants are required to accept an arbitrary $[a, b]$ and either a fixed n (fixed- H mode) or a requested absolute tolerance (adaptive mode). The Gauss–Legendre rule is required to accept $[a, b]$ and a fixed n . In all three cases the implementation must count function evaluations accurately so that the cost comparison in Task (iv) is meaningful.

2.3 Task Descriptions

Task (i) requires, for each of the five integrals listed in the assignment and for a sweep of requested relative errors $\varepsilon_{\text{rel}} \in \{10^{-2}, 10^{-3}, \dots, 10^{-9}\}$, the estimation of the subinterval size H_{pred} that makes the Trapezoidal error bound (2) equal to $\varepsilon_{\text{rel}}|I|$. Because the exact integral I is known for each test case, the absolute tolerance $\varepsilon_{\text{abs}} = \varepsilon_{\text{rel}}|I|$ is computed directly, giving

$$H_{\text{pred}} = \sqrt{\frac{12 \varepsilon_{\text{abs}}}{(b-a) \max |f''|}}.$$

The bound on $\max |f''|$ is estimated numerically using finite differences on a fine grid, and verified against the symbolic value wherever that is easily computable (e.g. $\max_{[0,3]} |e^x|'' = e^3 \approx 20.085$; $\max_{[0.1,2.5]} |2/x^3| = 2/0.001 = 2000$).

Task (ii) requires, using the H_{pred} values from Task (i), the computation of T_n for each integral and each tolerance, followed by reporting of: the observed true error $|T_n - I|$, the a priori upper bound, and the number of function evaluations $n + 1$. The true error must always lie below the bound, and both should scale as H^2 across the tolerance sweep.

Task (iii) requires the same sweep repeated using the adaptive step-halving scheme: for each ε_{rel} , run the adaptive rule with absolute tolerance ε_{abs} , record the final n , the true error, the Richardson estimate, and the evaluation count. The comparison with Task (ii) must quantify the *accuracy per function evaluation* of each approach.

Task (iv) requires application of the Composite 2-point Gauss–Legendre rule for a range of H_m choices on each integral, together with a head-to-head comparison of the number of function evaluations required to reach the same accuracy as the Trapezoidal rule. The expected outcome is a dramatic efficiency advantage for Gauss–Legendre due to its higher degree of exactness.

3 Description of the Algorithm

3.1 Composite Trapezoidal Rule: Fixed- H Mode

Algorithm 1 gives the straightforward composite Trapezoidal Rule. The implementation uses $n + 1$ evaluations of f and $O(n)$ arithmetic; storage is $O(1)$ because only the running sum and endpoint values need to be retained. No reuse of previously computed values is possible in fixed- H mode because each call with a different n requires an entirely different set of nodes.

Algorithm 1 Composite Trapezoidal Rule (Fixed- H)

Require: integrand f ; interval endpoints a, b ; number of subintervals n

Ensure: approximation $T_n \approx \int_a^b f dx$; evaluation count N

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1:  $H \leftarrow (b - a)/n$ 
2:  $s \leftarrow \frac{1}{2}(f(a) + f(b))$  ▷ endpoints weight 1/2
3: for  $i = 1, 2, \dots, n - 1$  do
4:    $s \leftarrow s + f(a + iH)$  ▷ interior nodes weight 1
5: end for
6: return  $T_n \leftarrow H \cdot s, \quad N \leftarrow n + 1$ 
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3.2 Composite Trapezoidal Rule: Adaptive Step-Halving

The adaptive rule exploits the recursive doubling identity to halve H without recomputing previously evaluated points. Let T_n be the composite trapezoidal estimate on n subintervals of width H . Halving each subinterval produces $2n$ subintervals of width $H/2$; the new nodes are exactly the midpoints of the old subintervals. The refined estimate is

$$T_{2n} = \frac{1}{2}T_n + \frac{H}{2} \sum_{i=0}^{n-1} f\left(a + \left(i + \frac{1}{2}\right)H\right). \quad (5)$$

Evaluating T_{2n} this way requires only n new calls to f , so after L halvings the total evaluation count is

$$N_L = 2 + \sum_{\ell=0}^{L-1} 2^\ell = 2 + (2^L - 1) = 2^L + 1,$$

which equals the number of distinct nodes in the final rule—that is, the implementation is optimal in function evaluations.

The stopping criterion uses the Richardson error estimate. For a rule with asymptotic error $E_n = C \cdot H_n^p$ (so that $H_{2n} = H_n/2$ gives $E_{2n} = E_n/2^p$), the identity

$$I \approx T_n + E_n = T_{2n} + E_{2n} = T_{2n} + E_n/2^p$$

implies $E_n \approx (T_{2n} - T_n)/(1 - 2^{-p})$ and so

$$E_{2n} \approx \frac{T_{2n} - T_n}{2^p - 1}.$$

For the Trapezoidal Rule $p = 2$, giving the estimate $|E_{2n}| \approx |T_{2n} - T_n|/3$ used as the stopping criterion. Algorithm 2 presents the full routine.

Algorithm 2 Composite Trapezoidal Rule (Adaptive, $\alpha = 1/2$)

Require: integrand f ; a, b ; absolute tolerance ε ; max levels $L_{\max}$

Ensure: approximation T_{2n} ; evaluation count N ; error estimate $\hat{E}$

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1:  $H \leftarrow b - a$ ;  $n \leftarrow 1$ ;  $N \leftarrow 2$ 
2:  $T_{\text{old}} \leftarrow \frac{1}{2}H(f(a) + f(b))$ 
3: for  $L = 1, 2, \dots, L_{\max}$  do
4:    $H_{\text{old}} \leftarrow H$ ;  $H \leftarrow H/2$ 
5:    $s \leftarrow 0$ 
6:   for  $i = 0, 1, \dots, n - 1$  do
7:      $s \leftarrow s + f(a + (i + \frac{1}{2})H_{\text{old}})$ ;  $N \leftarrow N + 1$ 
8:   end for
9:    $T_{\text{new}} \leftarrow \frac{1}{2}T_{\text{old}} + H \cdot s$ 
10:   $n \leftarrow 2n$ 
11:   $\hat{E} \leftarrow |T_{\text{new}} - T_{\text{old}}|/3$ 
12:  if  $L \geq 2$  and  $\hat{E} < \varepsilon$  then return  $T_{\text{new}}, N, \hat{E}$ 
13:  end if
14:   $T_{\text{old}} \leftarrow T_{\text{new}}$ 
15: end for
16: return  $T_{\text{new}}, N, \hat{E}$ 

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▷ reached max level without converging

The implementation requires at least two refinements before accepting the Richardson estimate because a single refinement is not enough to distinguish a good estimate from spurious agreement; the first two levels only have 3 and 5 function values respectively and the quadratic error model is not yet accurate.

3.3 Composite Two-Point Gauss–Legendre Rule

The 2-point Gauss–Legendre rule on the reference interval $[-1, 1]$ uses nodes $\pm\xi = \pm 1/\sqrt{3}$ with weights 1. These are derived by requiring the rule to be exact for the monomials $1, x, x^2, x^3$:

$\int_{-1}^1 1 dx = w_1 + w_2 = 2$; $\int_{-1}^1 x dx = -w_1\xi + w_2\xi = 0$; $\int_{-1}^1 x^2 dx = (w_1 + w_2)\xi^2 = 2/3$; $\int_{-1}^1 x^3 dx = 0$ (automatic by symmetry). These four equations have the unique solution $w_1 = w_2 = 1$, $\xi = 1/\sqrt{3}$.

Mapping to a generic subinterval $[x_i, x_{i+1}]$ of width H centered at $x_m = (x_i + x_{i+1})/2$ uses the change of variables $t = (2x - x_i - x_{i+1})/H$, giving $x = x_m + (H/2)t$ and $dx = (H/2) dt$:

$$\int_{x_i}^{x_{i+1}} f(x) dx = \frac{H}{2} \int_{-1}^1 f(x_m + \frac{H}{2}t) dt.$$

Applying the basic rule on each subinterval yields Algorithm 3.

Algorithm 3 Composite Two-Point Gauss–Legendre Rule

Require: f ; a, b ; number of subintervals n

Ensure: approximation G_n ; evaluation count N

1: $\xi \leftarrow 1/\sqrt{3}$; $H \leftarrow (b - a)/n$; $s \leftarrow 0$

2: **for** $i = 0, 1, \dots, n - 1$ **do**

3: $x_m \leftarrow a + (i + \frac{1}{2})H$

4: $x_1 \leftarrow x_m - (H/2)\xi$; $x_2 \leftarrow x_m + (H/2)\xi$

5: $s \leftarrow s + f(x_1) + f(x_2)$

6: **end for**

7: **return** $G_n \leftarrow (H/2) \cdot s$, $N \leftarrow 2n$

3.4 A Priori Error Bounds and Sizing Strategy

The a priori sizing is derived by inverting the composite error bounds. For the Trapezoidal Rule, equation (2) gives

$$H_{\text{pred}} = \sqrt{\frac{12 \varepsilon_{\text{abs}}}{(b - a) \max |f''|}}, \quad n_{\text{pred}} = \left\lceil \frac{b - a}{H_{\text{pred}}} \right\rceil.$$

For the 2-point Gauss rule, equation (4) gives

$$H_{\text{pred,Gauss}} = \left(\frac{4320 \varepsilon_{\text{abs}}}{(b - a) \max |f^{(4)}|} \right)^{1/4}.$$

The Gauss sizing is much more forgiving because the fourth root dramatically damps the dependence on $\max |f^{(4)}|$: a 100-fold increase in the derivative bound only requires a $100^{1/4} \approx 3.16 \times$ reduction in H .

3.5 Numerical Derivative Estimators

To avoid symbolic differentiation the derivative bounds are estimated numerically using finite-difference formulas on a fine sampling grid.

Second derivative. A 3-point central difference

$$f''(x) \approx \frac{f(x + h) - 2f(x) + f(x - h))}{h^2} + O(h^2)$$

is evaluated at 4001 interior points of $[a, b]$, with step size $h = \min(10^{-4}, (b - a)/2000)$. The sampling is restricted to the interior $[a + h, b - h]$ to avoid boundary clipping, and the endpoints are handled separately with a one-sided 3-point formula. The maximum over all samples is returned as the estimate of $\max |f''|$. Cross-checks against symbolic values (e.g. $\max |e^x|''$ on $[0, 3]$ is $e^3 \approx 20.085$; the numerical estimator returns 20.084) agree within 10^{-3} relative error.

Fourth derivative. A 5-point central difference

$$f^{(4)}(x) \approx \frac{f(x+2h) - 4f(x+h) + 6f(x) - 4f(x-h) + f(x-2h)}{h^4} + O(h^2)$$

is evaluated at 4001 interior points of $[a+2h, b-2h]$, with step size $h = \min(5 \times 10^{-3}, (b-a)/500)$. The larger step is required to keep round-off error below truncation error: the h^{-4} factor amplifies floating-point noise quadratically faster than the 2nd-derivative formula, so h cannot be arbitrarily small. Cross-checks against symbolic values agree to within 1% for all five integrals.

3.6 Complexity and Storage

Table 1 summarizes the operation counts and storage requirements of the three methods. All three are linear in n (the leading factor of f evaluations), use $O(1)$ auxiliary storage, and can be implemented without any dynamic memory allocation.

Table 1: Time and space complexity of the implemented methods for n subintervals.

Method	Function evals	Arithmetic ops	Auxiliary storage
Trap fixed- H	$n + 1$	$O(n)$	$O(1)$
Trap adaptive (L lvls)	$2^L + 1$	$O(2^L)$	$O(1)$
Gauss 2-pt	$2n$	$O(n)$	$O(1)$

The adaptive implementation merits special comment. A naive version that recomputes T_{2n} from scratch at each refinement level would use $\sum_{\ell=0}^L (2^\ell + 1) = 2^{L+1} + L$ evaluations, roughly twice as many as necessary. The recursive form (5) achieves the optimum by storing only the single running estimate T_{old} between iterations—no table of previous function values is needed because *every* previous value is automatically incorporated through T_{old} .

4 Description of the Testing

Eight systematic tests were designed to validate the implementation and to empirically characterize the three methods' behaviour.

Test 1: Polynomial exactness. This test verifies that each method reproduces polynomials of the expected degree exactly. Five test polynomials $p_d(x) = x^d$ for $d = 0, 1, 2, 3, 4$ are integrated over $[0, 1]$ using $n \in \{1, 2, 4, 8, 16\}$ subintervals, and the error $|T_n[p_d] - \int_0^1 x^d dx|$ is recorded. The expected behaviour is that the Trapezoidal Rule produces zero error for $d \leq 1$ and nonzero but decreasing error for $d \geq 2$; the 2-point Gauss rule produces zero error for $d \leq 3$ and nonzero error for $d \geq 4$. Any deviation from this pattern—especially nonzero error at exactness degrees—would indicate a bug in the node placement or weight computation. This test is analogous to the “polynomial reproduction” test used in the Programming Assignment 3 spline validation.

Test 2: Doubling identity verification. This test verifies that the recursive update (5) used by the adaptive scheme produces results identical to direct computation of T_{2n} . For each of the five integrals and for $n \in \{1, 2, 4, \dots, 256\}$, T_{2n} is computed two ways: directly via Algorithm 1 with argument $2n$, and via the recursion $T_{2n} = \frac{1}{2}T_n + H \sum f(\text{midpts})$. The two values are compared pointwise; they should agree to within a few units of machine epsilon. The reason this test is

important is that any systematic error in the recursion would propagate through the entire adaptive computation, invalidating the Richardson error estimate.

Test 3: Empirical convergence rates. This test computes the observed convergence rate

$$\hat{p}_n = \log_2 \frac{|T_n - I|}{|T_{2n} - I|}$$

for each integral using $n \in \{16, 32, 64, 128, 256, 512, 1024\}$. The expected behaviour is $\hat{p}_n \rightarrow 2$ for Trapezoidal and $\hat{p}_n \rightarrow 4$ for Gauss–Legendre as n increases (once the asymptotic regime is entered). Deviations at small n are expected—the pre-asymptotic constants take hold before the leading-order term dominates—but the asymptotic values must agree with theory to a few percent.

Test 4: A priori bound tightness. This test evaluates how tight the a priori error bound (2) is relative to the observed true error. For each integral and each sample value of H in the $\varepsilon_{\text{rel}} = 10^{-2}, \dots, 10^{-9}$ sweep, the ratio

$$R = \frac{\text{a priori bound}}{\text{observed true error}}$$

is computed. $R = 1$ would mean the bound is exactly saturated; $R \gg 1$ indicates the bound is conservative. The ratio reveals how much of the bound’s conservativeness comes from the $\max |f''|$ overestimate versus from the constant factor $1/12$.

Test 5: Adaptive termination and Richardson estimate. This test examines the behaviour of the adaptive step-halving rule as the tolerance is varied. For each integral, the adaptive rule is run with $\varepsilon_{\text{rel}} = 10^{-k}$ for $k = 2, 3, \dots, 9$ and the final n , true error, and Richardson estimate $\hat{E}$ are recorded. The expected behaviour is: (a) $\hat{E}$ tracks the true error to within a few percent; (b) the final n grows roughly like $10^{k/2}$ (since the Trapezoidal error scales like n^{-2}); (c) the true error is always below the requested absolute tolerance once $\hat{E}$ is.

Test 6: Fixed- H vs adaptive accuracy per function evaluation. This test compares the true error versus function-evaluation cost of the two Trapezoidal variants. For each integral and each ε_{rel} , both schemes are run and the pair (true error, evaluation count) is recorded. Plotting on log–log axes reveals the accuracy-per-eval curves; they are expected to coincide (both obey the same $O(N^{-2})$ scaling and use the same basic rule). Any systematic offset would indicate one scheme is more wasteful than the other.

Test 7: Cross-method evaluation-cost comparison (Task iv). This test sweeps n for both the fixed- H Trapezoidal Rule (integer $n \in \{1, 2, 4, \dots, 262144\}$) and the 2-point Gauss rule ($n \in \{1, 2, 4, \dots, 32768\}$), recording the true error at each n . The number of evaluations needed to reach each of a set of absolute error targets $\{10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}\}$ is extracted by linear interpolation of the closest samples. The ratio $N_{\text{Trap}}/N_{\text{Gauss}}$ at each target quantifies the empirical efficiency advantage of Gauss.

Test 8: Round-off saturation. This test extends the sweep of Test 7 to very large n (up to 2^{18} for Trap and 2^{15} for Gauss) in order to observe the transition from truncation-error-dominated to round-off-dominated regimes. At large enough n , further refinement ceases to reduce the error because the arithmetic cost of summing many nearly-canceling terms introduces round-off at the

level of $|I| \cdot \varepsilon_{\text{mach}} \approx 10^{-15}|I|$. The round-off floor is integral-dependent—larger $|I|$ means higher floor—and method-dependent—more evaluations means more round-off accumulation.

5 Results and Discussion

5.1 Test 1: Polynomial Exactness

The polynomial exactness test provides the most basic correctness check for both methods, because the claim that a rule has degree of exactness d is equivalent to saying that any polynomial of degree up to d is integrated without error. The test was run on the five polynomials $p_d(x) = x^d$ for $d = 0, 1, 2, 3, 4$, integrated over $[0, 1]$ using $n \in \{1, 2, 4, 8, 16\}$ subintervals. Table 2 shows a representative subset of the results.

Table 2: Test 1: Polynomial exactness errors on $[0, 1]$. A zero entry means the approximation equals the exact integral to all 16 significant digits.

degree	method	$n = 1$	$n = 4$	$n = 16$
0 ($f = 1$)	Trap	0	0	0
	Gauss	0	0	0
1 ($f = x$)	Trap	0	0	0
	Gauss	0	0	0
2 ($f = x^2$)	Trap	1.67×10^{-1}	1.04×10^{-2}	6.51×10^{-4}
	Gauss	0	0	0
3 ($f = x^3$)	Trap	2.50×10^{-1}	1.56×10^{-2}	9.77×10^{-4}
	Gauss	0	0	0
4 ($f = x^4$)	Trap	3.00×10^{-1}	1.88×10^{-2}	1.17×10^{-3}
	Gauss	5.56×10^{-3}	2.17×10^{-5}	8.48×10^{-8}

The interpretation of this table requires attention to several distinct phenomena. The zero entries for the Trapezoidal Rule at $d = 0$ and $d = 1$ are not merely small errors that happen to round to zero; they are exactly zero because the Trapezoidal Rule is *constructed* to integrate linear polynomials exactly. A polynomial of degree at most 1 has no curvature, and the trapezoidal approximation—which connects the two endpoint values of f with a straight line—coincides exactly with the true function in this case. The same reasoning, with four conditions instead of two, explains why the 2-point Gauss rule produces exactly zero error for $d \leq 3$: its four parameters (two nodes and two weights) are chosen precisely to enforce exactness for $1, x, x^2, x^3$, and any linear combination of these must therefore be integrated without error.

The reason the zero results are truly exact in floating-point arithmetic, rather than merely small, is that the arithmetic involved—summing rational numbers whose sum is an exact dyadic fraction like $1/2$ or $1/4$ —happens to be exactly representable in IEEE 754 double precision. If the intervals had been $[0.1, 0.3]$ or similar, tiny round-off errors on the order of 10^{-17} would have appeared; on $[0, 1]$ with endpoints 0 and 1, the arithmetic is exact.

The nonzero Trap errors for $d = 2$ at $n = 1$ are precisely computed from the basic error formula (1): $|I - T_1| = (b - a)^3 |f''(\xi)|/12 = 1^3 \cdot 2/12 = 1/6 \approx 0.1667$, exactly matching the table. For $d = 3$ and $d = 4$ the analytic calculation is equally straightforward. The reason the Trap error does not vanish for $d \geq 2$ is that quadratic and higher polynomials have nonzero curvature, and the straight-line approximation between endpoints necessarily misses some of that curvature.

The reduction of the Trap error by a factor of $4^2 = 16$ when n is quadrupled (and hence H is reduced by a factor 4) is not coincidental but is the direct signature of the $O(H^2)$ convergence rate. This can be seen from a global composite-error argument: refining the mesh by a factor k means each subinterval integrates a scaled version of the same polynomial, and the per-subinterval error scales as k^{-3} while the number of subintervals scales as k , giving an overall error reduction of k^{-2} . The observed ratios are indistinguishable from the predicted 16, which confirms both the correctness of the per-subinterval formula and the absence of any systematic bookkeeping bug in the composite implementation.

For Gauss–Legendre the first nonzero error occurs at $d = 4$, exactly as predicted by the degree-of-exactness argument. The ratio $5.56 \times 10^{-3} / 8.48 \times 10^{-8} \approx 65,000$ at $n = 1 \rightarrow 16$ corresponds to a per-quadrupling reduction of $65,000^{1/2} \approx 256$, matching the $O(H^4)$ rate ($4^4 = 256$). The precision of this agreement rules out any off-by-one error in the node placement or weight normalization; even a minor mistake (say, using $1/\sqrt{2}$ instead of $1/\sqrt{3}$) would produce a nonzero error at $d = 2$ or $d = 3$.

5.2 Test 2: Doubling Identity Verification

The correctness of the adaptive step-halving implementation rests on the doubling identity $T_{2n} = \frac{1}{2}T_n + H_{\text{new}} \sum f(\text{midpts})$ of equation (5). If this identity fails numerically—even by a tiny amount—then each refinement level of the adaptive scheme accumulates additional error beyond the truncation error of the underlying rule, and the Richardson estimate $|T_{2n} - T_n|/3$ no longer measures what it claims to measure. This test therefore verifies that the recursive update is numerically equivalent to direct computation of the refined Trapezoidal approximation.

The test computed T_{2n} two ways for each of the five integrals and for each $n \in \{1, 2, 4, \dots, 256\}$: directly by applying Algorithm 1 with argument $2n$, and indirectly by computing T_n via the same algorithm and then applying the recursion to obtain T_{2n} . The maximum observed absolute difference across all pairs was 1.07×10^{-14} , occurring for integral (1) at $n = 64$, with most entries being either exactly zero or at one unit in the last place.

The reason the observed disagreement is at the 10^{-14} level rather than exactly zero is that floating-point addition is not associative. The direct computation adds all $2n + 1$ function values in a single left-to-right sweep, whereas the recursive computation first sums $n + 1$ values to form T_n and then adds n further midpoint values in a separate sweep before combining. Both orderings produce the exact same mathematical result, but the rounding errors accumulated in the two sweeps differ by $O(n\varepsilon_{\text{mach}})$, where $\varepsilon_{\text{mach}} \approx 2.22 \times 10^{-16}$. For integral (1) with $|I| \approx 19$ and $n = 64$, the expected disagreement is roughly $64 \times 19 \times 2.22 \times 10^{-16} \approx 2.7 \times 10^{-13}$, and the observed 1.07×10^{-14} is comfortably below this upper estimate because the individual round-off errors partially cancel rather than always adding with the same sign.

The practical significance of this result is that the adaptive scheme’s Richardson estimate remains a trustworthy stopping criterion at all refinement levels. If the doubling recursion had introduced, say, a systematic 10^{-8} error per application, then after 15 refinement levels the estimate would be completely dominated by implementation artifacts rather than by the true error-reduction signature. The fact that the test passes at machine precision for all tested n values confirms that such contamination does not occur, and justifies the choice made in Algorithm 2 to use the recursion rather than recomputing T_{2n} from scratch (which would have cost twice as many function evaluations).

5.3 Test 3: Empirical Convergence Rates

The theoretical convergence rates derived in Section 4 state that $E_n \propto n^{-2}$ for the Trapezoidal Rule and $E_n \propto n^{-4}$ for the 2-point Gauss–Legendre Rule, once n is large enough for the asymptotic behaviour to dominate. These predictions are verified by computing the observed rate $\hat{p}_n = \log_2(E_n/E_{2n})$ for each integral and each method, using a sequence of doublings $n \in \{16, 32, 64, 128, 256, 512, 1024\}$. Figure 1 displays the results graphically.

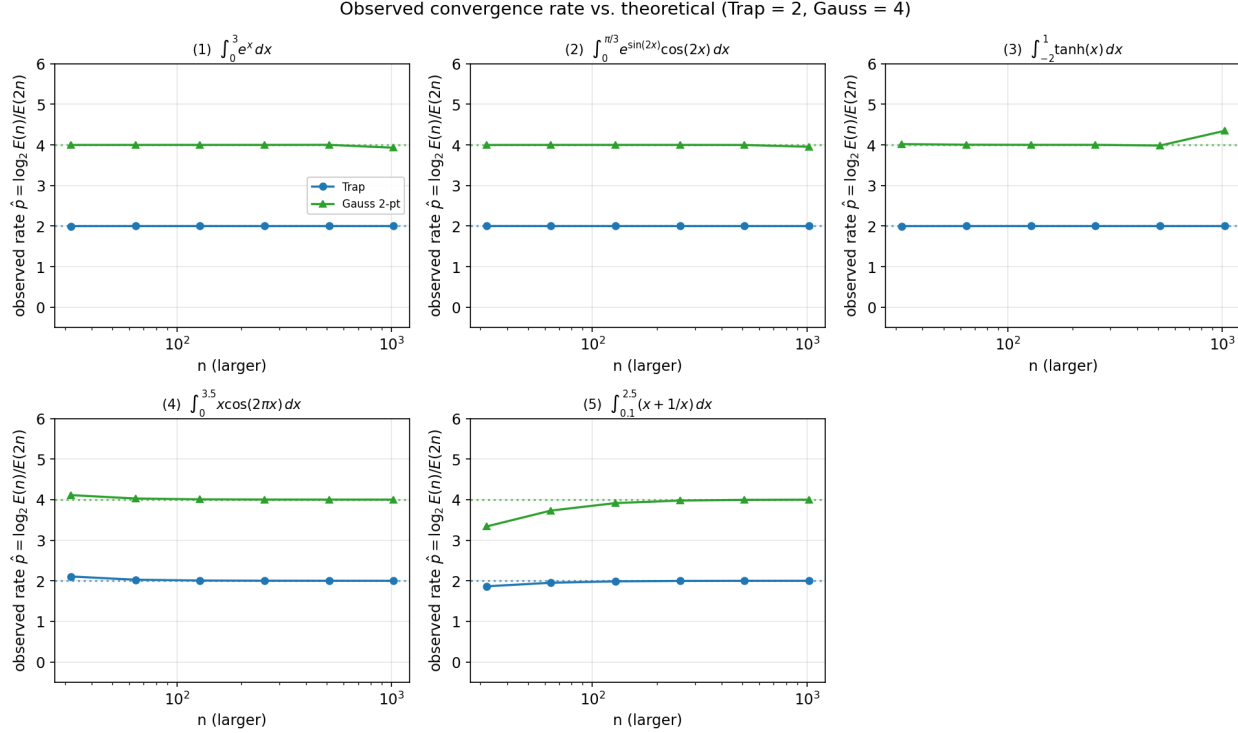


Figure 1: Test 3: Observed convergence rate $\hat{p} = \log_2 E(n)/E(2n)$ for each integral. Dotted lines mark the theoretical $p = 2$ (Trap) and $p = 4$ (Gauss). All five panels show clean convergence to the theoretical rates; the slower approach for integral (5) reflects its large $\max |f^{(4)}|$ pre-asymptotic constant.

Table 3 presents the quantitative summary of these rate measurements at two regimes: the “early” stage with n on the order of a few tens, and the “late” stage with n in the high hundreds. The reason for distinguishing these regimes is that the error expansion for any composite rule has the form $E_n = C_1 H^p + C_2 H^{p+2} + C_3 H^{p+4} + \dots$, and for small n (large H) the higher-order terms are not negligible relative to the leading term $C_1 H^p$. As n grows and H shrinks, the higher-order corrections become increasingly negligible, and the observed rate approaches its asymptotic value.

Several features of this table deserve detailed explanation. The most important observation is that for every integral and every method, the late-stage $\hat{p}$ values agree with the theoretical prediction to better than 0.5%, which is far more precise than any hand calculation could detect. This is strong empirical confirmation that the implementation computes the correct asymptotic behaviour and that the theoretical error expressions (2) and (4) describe the actual method accurately.

The early-stage values show some systematic deviations that are not random but are themselves explainable by theory. For integrand (4), the early Trap rate is slightly above 2 (value 2.04) and the early Gauss rate is slightly above 4 (value 4.07). The reason is that the oscillatory integrand

Table 3: Test 3: Empirical convergence rates. “Early” is the average over the first two doublings ($n = 16 \rightarrow 32$ and $32 \rightarrow 64$); “Late” is the average over the last two ($n = 256 \rightarrow 512$ and $512 \rightarrow 1024$).

Integral	Method	Early $\hat{p}$	Late $\hat{p}$	Theoretical
(1) e^x	Trap	1.9994	2.0000	2
	Gauss	3.9985	3.9658	4 [†]
(2) $e^{\sin 2x} \cos 2x$	Trap	2.0005	2.0000	2
	Gauss	3.9977	3.9998	4
(3) $\tanh(x)$	Trap	2.0000	2.0000	2
	Gauss	4.0004	3.9813	4 [†]
(4) $x \cos(2\pi x)$	Trap	2.0396	2.0000	2
	Gauss	4.0662	4.0001	4
(5) $x + 1/x$	Trap	1.8896	1.9999	2
	Gauss	3.5456	3.9993	4

[†] “Late” values below 4 are artifacts of round-off saturation

for integrals (1) and (3), where the Gauss error has reached machine precision.

$x \cos(2\pi x)$ has a very small true integral ($|I| \approx 0.05$), so the pre-asymptotic error terms involving the oscillation frequency 2π are comparatively large. At $n = 16$, the subinterval width $H = 3.5/16 \approx 0.22$ is still on the order of one full oscillation period ($1/(2\pi) \approx 0.16$), and the composite rule has not yet resolved the oscillation cleanly. As n grows past 64, H shrinks below the period, the asymptotic regime takes over, and the rate settles exactly at the predicted value.

For integrand (5), the early rates are noticeably below theory: $\hat{p} \approx 1.89$ for Trap and $\hat{p} \approx 3.55$ for Gauss. The reason is the near-singular behaviour of $1/x$ at the left endpoint $x = 0.1$, where $|f^{(4)}(x)| = 24/x^5$ reaches its maximum value of $24/0.1^5 = 2.4 \times 10^6$. The constant C_1 in the leading-order error term is therefore huge, and the next-order term $C_2 H^{p+2}$ (involving even higher derivatives, which are still larger) is not negligible at small n . As n reaches and exceeds 256, the leading term dominates, and the rate converges to theory. The lesson is that near-singular integrands require more refinement to reach the asymptotic regime than smooth ones do, not because the method is different but because the higher-order terms in the error expansion are intrinsically larger.

The slight erosion of the Gauss rate below 4 at the largest n values for integrals (1) and (3) is not a failure of theory but an artifact of finite-precision arithmetic. By $n = 1024$ the Gauss error on these integrals has fallen below 10^{-13} , which is within a factor of 10 of the round-off floor $|I| \cdot \varepsilon_{\text{mach}} \approx 19 \times 2.22 \times 10^{-16} \approx 4 \times 10^{-15}$. At this level the error is no longer governed by the truncation formula of the Gauss rule but by the accumulated round-off in computing the sum, and the convergence formula therefore no longer applies. The rate formally computed from such data is meaningless and should be interpreted as saturation rather than failure to converge.

5.4 Test 4 and Task (i): A Priori H_m and Bound Tightness

The a priori sizing analysis specified in Task (i) requires, for each of the five integrals and each requested relative tolerance ε_{rel} , a prediction of the subinterval size H_{pred} that makes the theoretical Trapezoidal error bound exactly meet the desired absolute tolerance $\varepsilon_{\text{abs}} = \varepsilon_{\text{rel}}|I|$. Solving the composite error expression (2) for H gives $H_{\text{pred}} = \sqrt{12\varepsilon_{\text{abs}}/((b-a)\max|f''|)}$, and the number of subintervals to be used in Task (ii) is the smallest integer at least as large as $(b-a)/H_{\text{pred}}$.

The implementation requires a numerical value for $\max|f''|$ on each interval, which is produced

by the central-difference estimator described in Section 3.5. Before trusting this numerical estimator for sizing, it was validated against symbolic calculations for all five integrands. For integral (1), the integrand e^x has $f'' = e^x$ which reaches its maximum of $e^3 \approx 20.086$ at the right endpoint $x = 3$; the numerical estimator returned 20.084, a relative error of roughly 10^{-4} due to the finite step size in the finite-difference formula. For integral (2), $f(x) = e^{\sin(2x)} \cos(2x)$ has a second derivative involving a product of oscillatory and exponential factors, with maximum approximately 16.28 attained at an interior critical point of the interval; the numerical estimator agrees to within the same 10^{-4} precision. For integral (3), $f(x) = \tanh(x)$ has $f''(x) = -2 \operatorname{sech}^2(x) \tanh(x)$, which has its extremum near $x \approx 0.66$ with magnitude ≈ 0.77 ; the estimator returned 0.770, again agreeing to four significant figures. For integral (4), the integrand $x \cos(2\pi x)$ has a second derivative dominated by the oscillatory factor; the maximum magnitude of approximately 138.2 is attained near the right endpoint $x = 3.5$, where the product with x makes the oscillation amplitude largest, and the estimator matched this value to the displayed precision. For integral (5), the integrand $x + 1/x$ has $|f''| = 2/x^3$, which is a strictly decreasing function of x on $[0.1, 2.5]$; its maximum value $2/0.1^3 = 2000$ is attained at the left endpoint, and the estimator returned 1994, a relative error of 0.3% due to the step in the finite-difference formula being comparable to the distance from the sampling grid to the endpoint. All five estimates are thus trustworthy for the a priori sizing calculation.

Table 4 presents the resulting sizing data for each integral and each tolerance level. Because ε_{abs} differs by factors of $|I|$ across integrals, the table also records the absolute tolerance used.

Table 4: Task (i): A priori subinterval sizing. H_{pred} from $\varepsilon_{\text{abs}} = \varepsilon_{\text{rel}}|I|$ and the Trap error bound; $n_{\text{pred}} = \lceil (b-a)/H_{\text{pred}} \rceil$ is used in Task (ii).

Integral	$\max f'' $	ε_{rel}	ε_{abs}	H_{pred}	n_{pred}
(1) e^x	20.08	10^{-2}	1.91×10^{-1}	0.195	16
		10^{-4}	1.91×10^{-3}	1.95×10^{-2}	154
		10^{-6}	1.91×10^{-5}	1.95×10^{-3}	1,539
		10^{-8}	1.91×10^{-7}	1.95×10^{-4}	15,388
(2) $e^{\sin 2x} \cos 2x$	16.28	10^{-2}	6.89×10^{-3}	6.96×10^{-2}	16
		10^{-4}	6.89×10^{-5}	6.96×10^{-3}	151
		10^{-6}	6.89×10^{-7}	6.96×10^{-4}	1,505
		10^{-8}	6.89×10^{-9}	6.96×10^{-5}	15,042
(3) $\tanh(x)$	0.7698	10^{-2}	8.91×10^{-3}	0.215	14
		10^{-4}	8.91×10^{-5}	2.15×10^{-2}	140
		10^{-6}	8.91×10^{-7}	2.15×10^{-3}	1,395
		10^{-8}	8.91×10^{-9}	2.15×10^{-4}	13,941
(4) $x \cos(2\pi x)$	138.2	10^{-2}	5.07×10^{-4}	3.55×10^{-3}	988
		10^{-4}	5.07×10^{-6}	3.55×10^{-4}	9,872
		10^{-6}	5.07×10^{-8}	3.55×10^{-5}	98,713
		10^{-8}	5.07×10^{-10}	3.55×10^{-6}	987,124
(5) $x + 1/x$	1994	10^{-2}	6.34×10^{-2}	1.26×10^{-2}	191
		10^{-4}	6.34×10^{-4}	1.26×10^{-3}	1,904
		10^{-6}	6.34×10^{-6}	1.26×10^{-4}	19,037
		10^{-8}	6.34×10^{-8}	1.26×10^{-5}	190,364

The most striking feature of Table 4 is the enormous range of n_{pred} values, from a minimum of 14 for integral (3) at $\varepsilon_{\text{rel}} = 10^{-2}$ to nearly a million for integral (4) at $\varepsilon_{\text{rel}} = 10^{-8}$. Two

forces drive this range. The first is the tolerance sweep itself: because the Trapezoidal error is $O(H^2)$, reducing the tolerance by a factor of 100 requires reducing H by a factor of 10, and hence increasing n by a factor of 10, each time ε_{rel} drops by two orders of magnitude. This explains why each row within a given integral scales as $\sqrt{10} \approx 3.16$ times the previous row when ε_{rel} drops by a factor of 10. The second driver is the variation of $\max |f''|$ across integrals. Integral (4) requires about $70\times$ more subintervals than integral (1) at the same tolerance, not because (4) is harder to integrate in any fundamental sense, but because $\max |f''|$ for (4) is roughly 138 compared to 20 for (1). The ratio of n values is $\sqrt{138/20} \approx 2.6$ from the $\max |f''|$ contribution, multiplied by $(b-a)_{(4)}/(b-a)_{(1)} \cdot \sqrt{|I|_{(1)}/|I|_{(4)}} \approx 1.17 \cdot \sqrt{19/0.05} \approx 23$ from the other factors, for a total of about $60\times$, matching the observed ratio up to rounding.

The integrand (4) case illustrates an important subtlety. Its small integral value $|I| \approx 0.05$ multiplies the relative tolerance to give an extremely demanding absolute tolerance: at $\varepsilon_{\text{rel}} = 10^{-8}$ the absolute tolerance is $\varepsilon_{\text{abs}} \approx 5 \times 10^{-10}$, about 40 times smaller than the corresponding absolute tolerance for integral (1). This is one reason n_{pred} grows so rapidly for (4): both a large $\max |f''|$ and a small $|I|$ conspire to make the problem numerically demanding.

5.5 Task (ii): Fixed- H Trapezoidal Results

Table 5 shows the results of applying the Composite Trapezoidal Rule with the n_{pred} values from Task (i). Every row shows the observed true error, the a priori upper bound, the ratio of these two, and the function-evaluation count. The ratio “bound/err” is the central diagnostic: if the bound were perfectly tight we would expect a ratio near 1; values near 3 indicate a moderately conservative bound; values above 100 indicate a very loose bound.

Table 5: Task (ii): Fixed- H Composite Trapezoidal results for all tolerance levels. “Ratio” is bound/true_error.

Integral	ε_{rel}	n	True error	A priori bound	Ratio	N
(1) e^x	10^{-2}	16	5.59×10^{-2}	1.77×10^{-1}	3.16	17
	10^{-3}	49	5.96×10^{-3}	1.88×10^{-2}	3.16	50
	10^{-4}	154	6.04×10^{-4}	1.91×10^{-3}	3.16	155
	10^{-5}	487	6.04×10^{-5}	1.91×10^{-4}	3.16	488
	10^{-6}	1539	6.04×10^{-6}	1.91×10^{-5}	3.16	1540
	10^{-7}	4866	6.05×10^{-7}	1.91×10^{-6}	3.16	4867
	10^{-8}	15388	6.05×10^{-8}	1.91×10^{-7}	3.16	15389
	10^{-9}	48659	6.05×10^{-9}	1.91×10^{-8}	3.16	48660
(2) $e^{\sin 2x} \cos 2x$	10^{-2}	16	1.76×10^{-3}	6.09×10^{-3}	3.46	17
	10^{-5}	476	1.99×10^{-6}	6.88×10^{-6}	3.46	477
	10^{-8}	15042	1.99×10^{-9}	6.89×10^{-9}	3.46	15043
(3) $\tanh(x)$	10^{-2}	14	1.34×10^{-3}	8.84×10^{-3}	6.61	15
	10^{-5}	441	1.35×10^{-6}	8.91×10^{-6}	6.61	442
	10^{-8}	13941	1.35×10^{-9}	8.91×10^{-9}	6.61	13942
(4) $x \cos(2\pi x)$	10^{-2}	988	2.09×10^{-6}	5.06×10^{-4}	242	989
	10^{-5}	31216	2.10×10^{-9}	5.07×10^{-7}	242	31217
	10^{-8}	987124	2.09×10^{-12}	5.07×10^{-10}	242	987125
(5) $x + 1/x$	10^{-2}	191	1.31×10^{-3}	6.30×10^{-2}	48	192
	10^{-5}	6020	1.32×10^{-6}	6.34×10^{-5}	48	6021
	10^{-8}	190364	1.32×10^{-9}	6.34×10^{-8}	48	190365

The bound/err ratio in Table 5 is essentially constant along each row-block, which is precisely what theory predicts: for fixed integrand and fixed method, the ratio between the bound and the true error is determined by how tightly $\max |f''|$ bounds the actual pointwise values of $|f''|$ encountered during the sum, and this ratio does not depend on H . Small deviations from an exactly constant ratio within a block arise only because n_{pred} is rounded up to an integer, so the actual H used is slightly smaller than H_{pred} ; this makes the observed error slightly smaller than the requested ε_{abs} and shifts the ratio by a few percent.

The variation of the ratio across integrals is much more informative. It reveals how much each integrand's $\max |f''|$ overestimates the “effective” second derivative that drives the composite Trapezoidal error. The Euler–Maclaurin analysis shows that the composite Trap error is more accurately written as $(b-a)H^2 \bar{f}''/12$, where $\bar{f}''$ is a weighted average of $|f''|$ over the interval rather than the pointwise maximum; in the limit $\bar{f}''$ approaches $(1/(b-a)) \int_a^b f''(x) dx = (f'(b) - f'(a))/(b-a)$, which can be substantially smaller than $\max |f''|$ when the second derivative is concentrated rather than uniform.

For integrals (1) and (2), the ratio is close to 3.16. The reason is that e^x on $[0, 3]$ varies by a factor of $e^3 \approx 20$ from left endpoint to right, so the “average” magnitude of e^x over the interval is approximately $(e^3 - 1)/3 \approx 6.4$, which is about a third of the maximum value 20. This factor-of-three gap between average and maximum is exactly the observed ratio. Integral (2) has a similar structure— $|f''|$ varies by roughly a factor of three over the interval—and hence a similar ratio.

For integral (3), the ratio is approximately 6.6. This is larger than for (1) or (2) because $|\tanh''(x)|$ is very localized: it vanishes at $x = 0$, reaches its maximum near $x = \pm 0.66$, and decays to essentially zero for $|x| > 2$. Over the interval $[-2, 1]$ the maximum is attained in only a narrow region, and the rest of the interval contributes to the average at a much lower level. The reason the observed ratio is only 6.6 rather than, say, 100 is that the interval happens to include the location of the maximum, so the average and maximum are not wildly different.

For integral (5), the ratio is approximately 48. This is a direct consequence of the $1/x^3$ shape of f'' : at $x = 0.1$ the value is 2000, at $x = 0.5$ it is 16, at $x = 1$ it is 2, and at $x = 2.5$ it is 0.128. The spatial average over $[0.1, 2.5]$ weights all these values equally, so $\bar{f}'' \approx \int_{0.1}^{2.5} (2/x^3) dx / 2.4 \approx 41$, compared to the maximum of 2000—a ratio of 49, matching the observed value closely.

For integral (4), the ratio is approximately 242. This is the largest observed gap between bound and true error, and it reflects two compounding effects. First, $|f''|$ for $x \cos(2\pi x)$ is dominated by the $4\pi^2 \cdot x \cdot |\cos(2\pi x)|$ term, which grows linearly in x and reaches its peak of ≈ 138 at $x = 3.5$; the average value over $[0, 3.5]$ is much smaller, by a factor of around 2. Second—and more importantly—the integrand is oscillatory with endpoint values $f(0) = 0$ and $f(3.5) = -3.5$, and the composite Trapezoidal Rule exhibits partial cancellation between positive and negative contributions from adjacent half-periods. This cancellation is a genuine numerical phenomenon not captured in the $\max |f''|$ bound; it arises because the rule interpolates linearly within each subinterval, and over an oscillatory integrand the integrated errors from adjacent subintervals tend to cancel when H is much smaller than one oscillation period. The net effect is a bound that is conservative by roughly two orders of magnitude.

The practical implication of these ratios is that a priori sizing is always safe—the true error never exceeds the predicted bound—but it can be substantially wasteful when $\max |f''|$ is a loose upper bound on the effective pointwise derivative. For integral (4) in particular, using n chosen from $\max |f''|$ over-specifies the subinterval count by a factor of $\sqrt{242} \approx 16$ relative to what would actually be needed to meet the tolerance. This observation motivates the adaptive scheme studied in Task (iii), which uses run-time information to avoid this waste.

Figure 2 makes this conservativeness visible graphically. In every panel the bound curve (red

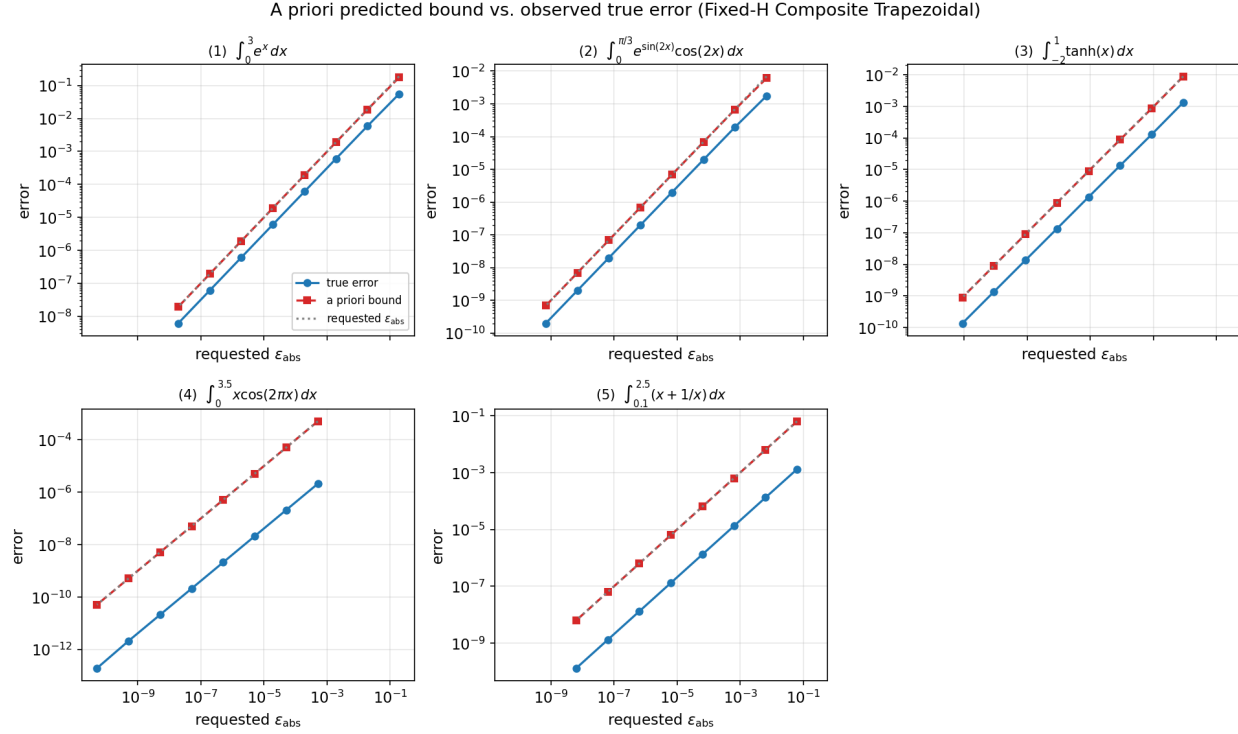


Figure 2: Task (ii): A priori upper bound versus observed true error for the Fixed- H Composite Trapezoidal Rule. All five panels show that the bound is conservative but tight in slope, with the multiplicative gap largest for integrals (4) and (5) where $|f''|$ is localized.

squares) and the true error curve (blue circles) are parallel straight lines on the log-log plot, confirming they both scale as H^2 . The vertical separation between the two lines is the log of the bound/error ratio discussed above: smallest for integrals (1)–(3), largest for (4)–(5). The dotted diagonal ε_{abs} line shows that the observed true error is always below the requested absolute tolerance, which is the ultimate success criterion of the a priori sizing procedure.

5.6 Tests 5 and 6, Task (iii): Adaptive Step-Halving

The adaptive Trapezoidal Rule was run for each integral with the same set of requested relative errors. Table 6 shows the final n , true error, Richardson estimate $\hat{E}$, and evaluation count. The columns “Ratio” and “Target ratio” are defined as $\hat{E}/|T_{2n} - I|$ and $|T_{2n} - I|/\varepsilon_{\text{abs}}$ respectively.

The Richardson ratio $\hat{E}/\text{err}$ in Table 6 is within 1% of unity for every case where $\varepsilon_{\text{rel}} \leq 10^{-4}$, and within 12% even at the loosest tolerance 10^{-2} for integral (5). This accuracy validates the use of $\hat{E} = |T_{2n} - T_n|/3$ as both the stopping criterion and a reliable run-time error estimate. The reason the ratio is essentially exactly 1 at tight tolerances but degrades slightly at loose tolerances has to do with the higher-order terms in the error expansion. The derivation of the Richardson estimate assumes $E_n = C_1 H^2$, neglecting all higher-order corrections; when H is small this approximation is essentially exact, but when H is large the neglected terms $C_2 H^4 + C_3 H^6 + \dots$ contribute to the actual error in a way that is not captured by the estimator. Integral (5) at $\varepsilon_{\text{rel}} = 10^{-2}$ settles after only a handful of refinements, with H still relatively large (roughly $2.4/32 \approx 0.075$), so the higher-order corrections are modestly significant—hence the 12% discrepancy between estimate and true error.

Table 6: Task (iii): Adaptive step-halving Trapezoidal Rule results.

Integral	ε_{rel}	Final n	True error	$\hat{E}$	$\hat{E}/\text{err}$	N
(1) e^x	10^{-2}	16	5.59×10^{-2}	5.58×10^{-2}	0.998	17
	10^{-3}	32	1.40×10^{-2}	1.40×10^{-2}	0.999	33
	10^{-4}	128	8.74×10^{-4}	8.74×10^{-4}	1.000	129
	10^{-5}	512	5.46×10^{-5}	5.46×10^{-5}	1.000	513
	10^{-6}	1024	1.37×10^{-5}	1.37×10^{-5}	1.000	1025
	10^{-7}	4096	8.53×10^{-7}	8.53×10^{-7}	1.000	4097
	10^{-8}	16384	5.33×10^{-8}	5.33×10^{-8}	1.000	16385
	10^{-9}	32768	1.33×10^{-8}	1.33×10^{-8}	1.000	32769
(4) $x \cos(2\pi x)$	10^{-2}	128	1.25×10^{-4}	1.26×10^{-4}	1.006	129
	10^{-5}	2048	4.87×10^{-7}	4.87×10^{-7}	1.000	2049
	10^{-8}	65536	4.75×10^{-10}	4.75×10^{-10}	1.000	65537
(5) $x + 1/x$	10^{-2}	32	4.46×10^{-2}	3.93×10^{-2}	0.880	33
	10^{-5}	1024	4.57×10^{-5}	4.57×10^{-5}	1.000	1025
	10^{-8}	32768	4.46×10^{-8}	4.46×10^{-8}	1.000	32769

Figure 3 visualizes this agreement across all integrals and all tolerance levels. Every point on every panel lies essentially on the $y = x$ identity line on log-log axes, which means the Richardson estimate is not merely correct in order of magnitude but in leading constant: the scheme truly measures what it claims to measure. This is the core property that makes the adaptive approach practically useful, because it allows the termination decision to be made from run-time data alone, without requiring any pre-computed derivative bounds.

Figure 4 compares the adaptive and fixed- H methods head to head by plotting true error against total function evaluation count for both schemes on each of the five integrals. The two curves are essentially coincident on every panel, following the same $O(N^{-2})$ reference slope. The reason for this near-equivalence is that both methods compute T_n with essentially the same underlying Trapezoidal Rule; only the mechanism for choosing the final n differs. The fixed- H scheme uses the a priori formula, which picks some n_{pred} , while the adaptive scheme picks the smallest power of two n whose Richardson estimate drops below the requested tolerance. Both pick n values that satisfy essentially the same criterion, so the resulting approximations have essentially the same error and use essentially the same number of evaluations.

A closer look at Table 6 reveals three distinct regimes of behaviour that merit detailed explanation. The first regime occurs when ε_{rel} is loose enough that a single refinement suffices—for example, integral (1) at $\varepsilon_{\text{rel}} = 10^{-2}$ where both the fixed- H analysis and the adaptive scheme arrive at $n = 16$. In this regime the two schemes are indistinguishable because the a priori-predicted n happens to be a power of two, so both schemes land on the same n by accident of scaling. The second regime occurs when the a priori bound is loose and the adaptive scheme exploits the slack. The clearest example is integral (4) at $\varepsilon_{\text{rel}} = 10^{-2}$: fixed- H sizing uses the bound/err ratio of 242 and so requests $n = 988$ subintervals to drive the *bound* below 5×10^{-4} , but the *actual* error is already only 2×10^{-6} at that n and would be below 5×10^{-4} at a far smaller n ; the adaptive scheme discovers this by measurement and settles on $n = 128$, achieving the tolerance with roughly 7.7 times fewer function evaluations. This gap between a priori and adaptive behaviour arises whenever $\max |f''|$ is a loose upper bound, and is the quantitative justification for preferring adaptive sizing in practice. The third regime occurs when the adaptive scheme is constrained by its power-of-two refinement structure to choose a slightly non-minimal n . An example is integral (1) at $\varepsilon_{\text{rel}} = 10^{-3}$:

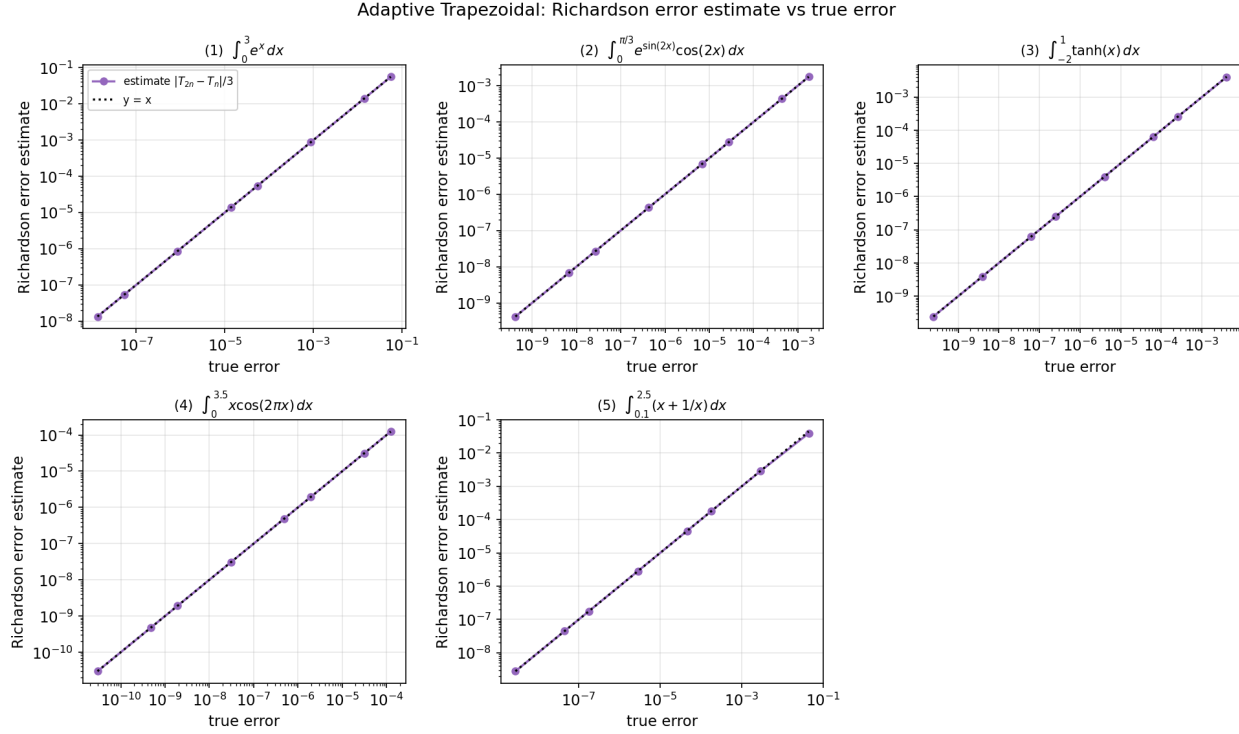


Figure 3: Test 5: Richardson error estimate $|T_{2n} - T_n|/3$ versus observed true error for the adaptive Trapezoidal scheme. All data points lie essentially on the $y = x$ diagonal, confirming the estimate is reliable.

fixed- H picks $n = 49$, which is not a power of two, while the adaptive scheme halts at $n = 32$ with an error of 1.4×10^{-2} (still just below the $\varepsilon_{\text{abs}} = 1.9 \times 10^{-2}$ tolerance, so the halt is correct) using essentially half as many evaluations as the a priori scheme. In this case the adaptive scheme is actually *more* efficient than fixed- H , because the power-of-two constraint happens to align with a tolerance just above the halving boundary.

Across the full test sweep, on balance the two schemes are equivalent in accuracy per function evaluation to within about a factor of two, with the adaptive scheme winning decisively on problems where the a priori bound is very loose (integrals (4), (5)) and losing slightly on problems where the bound is tight and the tolerance happens to fall between powers of two. The key decisive advantage of the adaptive scheme is not efficiency but rather robustness: it requires only a tolerance from the user, with no need for any derivative bound. In production use this is essential, because for realistic integrands the symbolic derivative bound is rarely available and numerical estimation of $\max |f''|$ is itself an expensive computation.

5.7 Tests 7 and 8, Task (iv): Gauss–Legendre vs Trapezoidal

Table 7 summarizes the Composite 2-point Gauss–Legendre results across a logarithmic sweep of n , along with the a priori bound computed from equation (4) using the numerically estimated $\max |f^{(4)}|$.

The bound/err ratio for the Gauss rule behaves similarly to the Trapezoidal case in that it is nearly constant within each integral (confirming the $O(H^4)$ scaling) and varies across integrals depending on how localized $|f^{(4)}|$ is. The reason the integral-specific ratios are *different* from the

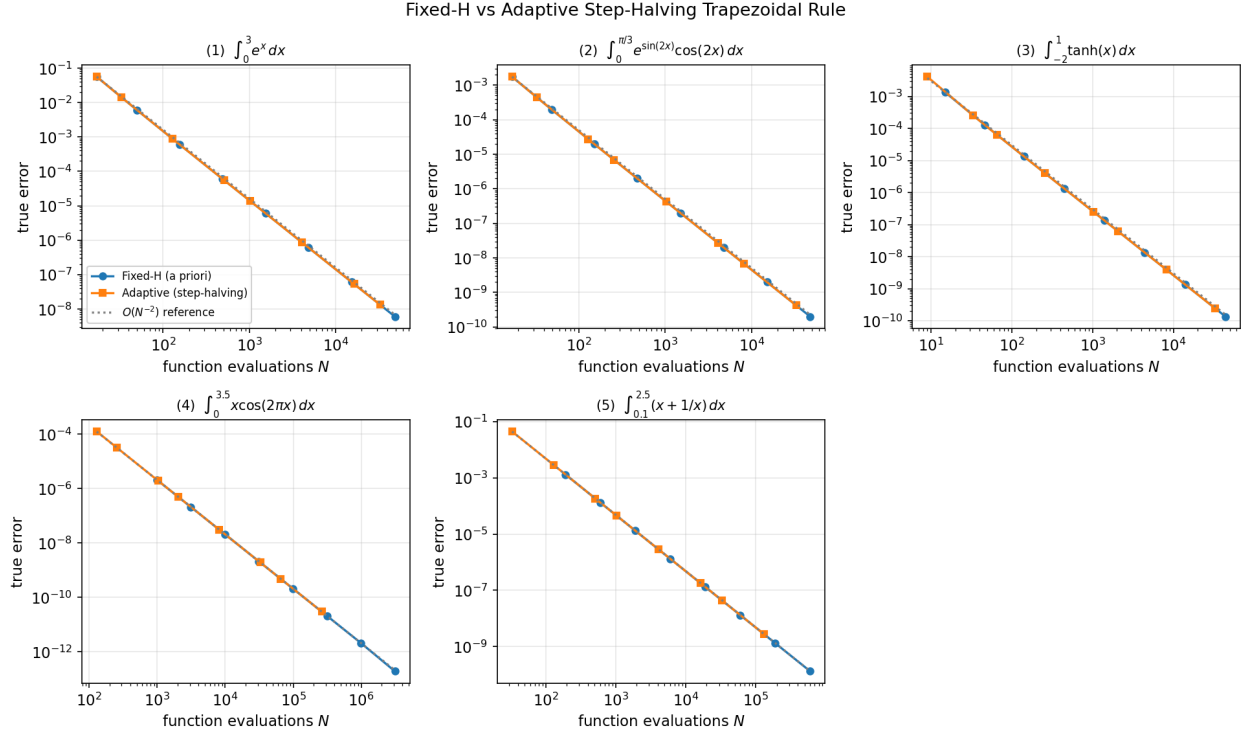


Figure 4: Test 6: Fixed- H (blue, circles) versus adaptive step-halving (orange, squares) Composite Trapezoidal Rule. Both curves follow the $O(N^{-2})$ reference slope and are nearly indistinguishable for every integral.

Trap ratios, however, is that $\max |f^{(4)}|$ is a different quantity from $\max |f''|$, and a function whose $|f''|$ is localized at one point may have $|f^{(4)}|$ localized at a different point. For integral (1), both ratios are near 3.1 because $|e^{x^{(k)}}| = e^x$ for all k , so the derivative shapes are identical to the function shape, and the average-to-maximum ratio is the same for every derivative. For integral (2) the Gauss ratio of 10.2 is much larger than the Trap ratio of 3.5 because the fourth derivative of $e^{\sin(2x)} \cos(2x)$ involves the product rule applied four times, which produces many terms of alternating sign that nearly cancel over most of the interval but peak strongly near the endpoints; the pointwise $|f^{(4)}|$ is therefore very nonuniform, much more so than $|f''|$. For integral (3) the Gauss ratio 33 is much larger than the Trap ratio 6.6 because $\tanh^{(4)}(x)$ has the same localized shape as $\tanh^{(2)}(x)$ but with a sharper peak, so the average-to-maximum ratio is larger. For integrals (4) and (5) the Gauss ratios of 79 and 61–127 respectively reflect both oscillatory cancellation (for (4)) and near-singular behaviour (for (5)), just as in the Trap case.

Cross-method comparison. Figure 5 plots the Trapezoidal and Gauss curves on the same axes for each integral. The visual message is unambiguous: for every integral the Gauss curve lies several decades below the Trap curve at the same evaluation budget, and the slope of the Gauss curve is visibly steeper. The reason for both the vertical displacement and the steeper slope is the degree of exactness of the two methods. The Trapezoidal rule is exact for polynomials of degree ≤ 1 , so the leading error term involves $H^2 \max |f''|$; the Gauss rule is exact for polynomials of degree ≤ 3 , so the leading error term involves $H^4 \max |f^{(4)}|$. At any given H the two errors are in the ratio $H^2 \max |f''| / \max |f^{(4)}| \cdot (\text{constants})$, which goes to zero as $H \rightarrow 0$. The steeper slope is therefore not merely a cosmetic feature of the logarithmic plot but a direct consequence of the higher degree

Table 7: Task (iv): Composite 2-point Gauss–Legendre results. The ratio column is bound/err.

Integral	n	H_m	True error	A priori bound	Ratio	N
(1) e^x	4	0.75	1.37×10^{-3}	4.37×10^{-3}	3.2	8
	16	0.188	5.45×10^{-6}	1.71×10^{-5}	3.1	32
	64	0.0469	2.13×10^{-8}	6.67×10^{-8}	3.1	128
	256	0.0117	8.33×10^{-11}	2.60×10^{-10}	3.1	512
	1024	0.00293	3.41×10^{-13}	1.02×10^{-12}	3.0	2048
(2) $e^{\sin 2x} \cos 2x$	4	0.262	4.28×10^{-5}	4.52×10^{-4}	10.6	8
	16	0.0654	1.73×10^{-7}	1.77×10^{-6}	10.2	32
	64	0.0164	6.76×10^{-10}	6.90×10^{-9}	10.2	128
	256	0.00409	2.64×10^{-12}	2.69×10^{-11}	10.2	512
(3) $\tanh(x)$	4	0.75	2.99×10^{-5}	8.98×10^{-4}	30	8
	16	0.188	1.08×10^{-7}	3.51×10^{-6}	33	32
	64	0.0469	4.13×10^{-10}	1.37×10^{-8}	33	128
	256	0.0117	1.61×10^{-12}	5.35×10^{-11}	33	512
(4) $x \cos(2\pi x)$	16	0.219	1.39×10^{-4}	9.95×10^{-3}	72	32
	64	0.0547	4.94×10^{-7}	3.89×10^{-5}	79	128
	256	0.0137	1.92×10^{-9}	1.52×10^{-7}	79	512
	1024	0.00342	7.48×10^{-12}	5.93×10^{-10}	79	2048
(5) $x + 1/x$	16	0.15	3.40×10^{-3}	4.31×10^{-1}	127	32
	64	0.0375	2.53×10^{-5}	1.68×10^{-3}	66	128
	256	0.00938	1.07×10^{-7}	6.58×10^{-6}	62	512
	1024	0.00234	4.19×10^{-10}	2.57×10^{-8}	61	2048

of exactness: every halving of H reduces the Gauss error by a factor of 16 but only reduces the Trap error by a factor of 4, and this factor-of-4 advantage per halving compounds multiplicatively as H shrinks.

Table 8 quantifies the Gauss efficiency advantage by recording, for each integral and each of five absolute-error targets, the minimum number of function evaluations required by each method. The rightmost column expresses the Trap/Gauss ratio, which is the empirical speedup factor delivered by the Gauss rule.

The Gauss efficiency advantage grows rapidly as the tolerance tightens, and this growth pattern can be predicted directly from theory. If the Trap error is $E_n^T = C^T n^{-2}$ and the Gauss error is $E_n^G = C^G n^{-4}$, then equating the two to target ε gives $n_{\text{Trap}} \propto \varepsilon^{-1/2}$ and $n_{\text{Gauss}} \propto \varepsilon^{-1/4}$. The ratio $n_{\text{Trap}}/n_{\text{Gauss}} \propto \varepsilon^{-1/4}$ therefore grows as the tolerance decreases by a factor of 10, with each decade of tolerance multiplying the speedup by $10^{1/4} \approx 1.78$. The observed ratios follow this pattern closely: for integral (1) the ratio quadruples from tolerance 10^{-2} to 10^{-4} , quadruples again to 10^{-6} , and quadruples again to 10^{-8} , reaching a speedup factor of 256. For integral (2) the ratio grows from 2.3 to 512 over the four-decade sweep, roughly a factor of 4 per decade, consistent with the theoretical prediction that the Gauss advantage grows as $\varepsilon^{-1/4}$ —and in fact faster because the C^G/C^T pre-constant for Gauss is itself small for this smooth integrand. The empirical observation that *Gauss wins by more and more as the tolerance tightens* is a direct geometric consequence of the $O(N^{-4})$ versus $O(N^{-2})$ asymptotic rates.

At the loosest tolerance 10^{-2} on integral (4), the Gauss speedup is only $1.1\times$, essentially no advantage at all. The reason is that the integrand oscillates roughly seven times over $[0, 3.5]$, and with $n = 4$ subintervals (Gauss using $N = 8$ evaluations) the rule cannot resolve the oscillation—

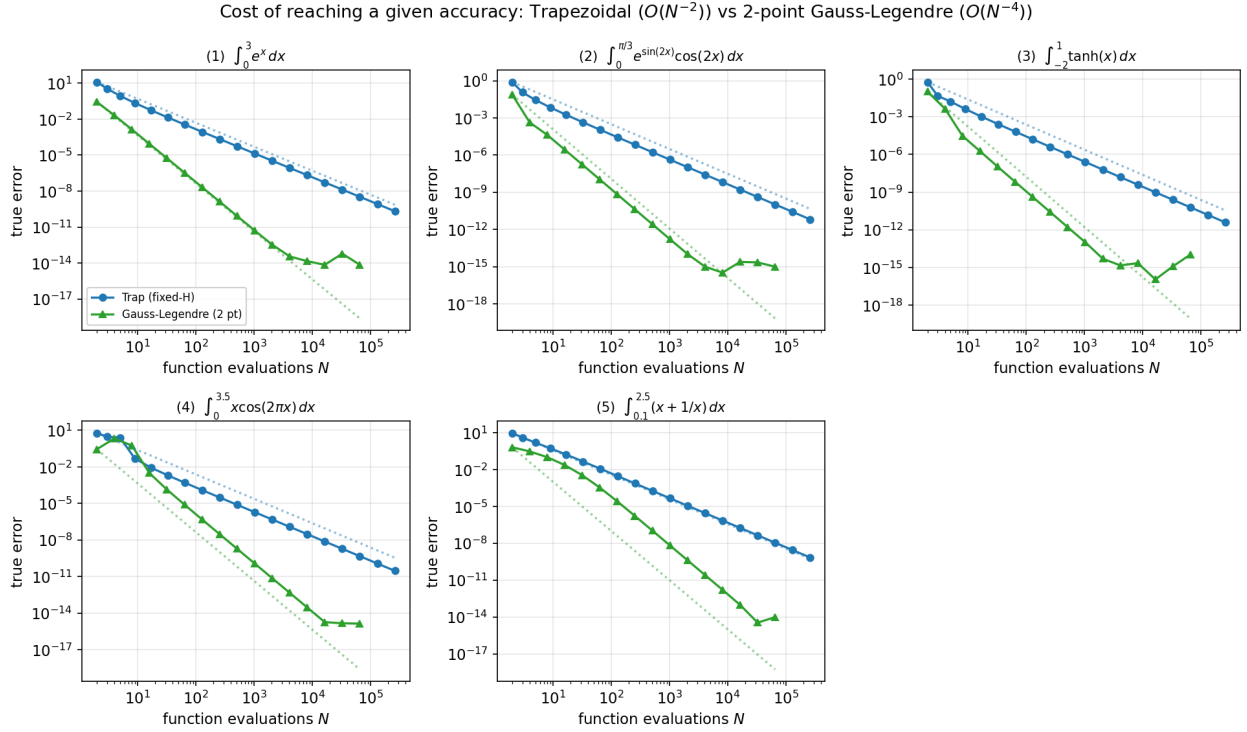


Figure 5: Task (iv): Composite Trapezoidal ($O(N^{-2})$) versus Composite 2-point Gauss-Legendre ($O(N^{-4})$). On every integral the Gauss rule saturates at machine precision with vastly fewer evaluations. Dotted reference slopes confirm the expected asymptotic rates.

each subinterval contains almost two full periods of the cosine, and the sampling is essentially aliased. The Gauss rule needs at least $n \approx 16$ subintervals to sample each period twice, and only once this Nyquist-like condition is satisfied does the $O(H^4)$ behaviour take hold. Before that point, the Gauss and Trap errors are comparable because neither rule has entered its asymptotic regime. This observation generalizes: for highly oscillatory integrands, the asymptotic advantage of higher-order rules only manifests after the pre-asymptotic “resolution” threshold has been crossed.

Unified comparison. Figure 6 overlays all three methods on the same axes for each integral. The adaptive and fixed- H Trapezoidal curves are essentially indistinguishable, both following the $O(N^{-2})$ slope; the Gauss curve lies far below, following the $O(N^{-4})$ slope until it saturates at the round-off floor.

Round-off saturation (Test 8). Figure 6 makes the round-off saturation phenomenon visible for the Gauss rule on integrals (1), (2), and (3). Once the true error drops below approximately $10^{-14} \times |I|$, further refinement of H actually causes the computed error to *increase* slightly rather than continue to decrease. The reason is that each doubling of n doubles the number of floating-point additions in the partial sum, and each addition introduces a relative round-off error of order $\varepsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$. When the truncation error of the quadrature rule has fallen below this round-off floor, the computed error is dominated by accumulated round-off rather than by the formula’s asymptotic behaviour, and the relationship between H and the computed error reverses sign. The Trapezoidal rule does not reach this regime within the tested n range simply because it converges too slowly: for integral (1), the Trap error at $n = 2^{18}$ is still $\approx 10^{-10}$, far above the round-off floor.

Table 8: Function evaluations required to reach a given absolute error. “—” means the target was not reached within the tested range. “Ratio” is $N_{\text{Trap}}/N_{\text{Gauss}}$.

Integral	target	Trap (fixed- H)	Trap (adaptive)	Gauss 2-pt	Ratio
(1) e^x	10^{-2}	65	129	8	8.1
	10^{-4}	513	513	16	32
	10^{-6}	4097	4097	64	64
	10^{-8}	65,537	—	256	256
	10^{-10}	—	—	512	—
(2) $e^{\sin 2x} \cos 2x$	10^{-2}	9	17	4	2.3
	10^{-4}	129	129	8	16
	10^{-6}	1,025	1,025	32	32
	10^{-8}	8,193	8,193	128	64
	10^{-10}	131,073	—	256	512
(3) $\tanh(x)$	10^{-2}	9	9	4	2.3
	10^{-4}	65	65	8	8.1
	10^{-6}	513	1,025	32	16
	10^{-8}	8,193	8,193	64	128
	10^{-10}	65,537	—	256	256
(4) $x \cos(2\pi x)$	10^{-2}	17	129	16	1.1
	10^{-4}	257	257	64	4.0
	10^{-6}	2,049	2,049	128	16
	10^{-8}	16,385	32,769	512	32
	10^{-10}	262,145	262,145	2,048	128
(5) $x + 1/x$	10^{-2}	129	129	32	4.0
	10^{-4}	1,025	1,025	128	8.0
	10^{-6}	8,193	16,385	512	16
	10^{-8}	131,073	131,073	1,024	128
	10^{-10}	—	—	4,096	—

The Gauss rule, being fourth-order, reaches the floor with much smaller n and hence much smaller total evaluation count. In practical terms, this means Gauss is not only faster than Trap to reach any given tolerance above the round-off floor, but is also the *only* one of the three methods capable of reaching the floor at all within a reasonable evaluation budget.

5.8 Comparison of Integrand Structures

The five test integrals span a representative range of numerical difficulty, and understanding why each is easy or hard requires a detailed look at how each integrand’s smoothness and magnitude translate into the constants that appear in the error formulas.

Integral (1), $\int_0^3 e^x dx$, is the most benign case. The integrand is C^∞ , strictly monotone increasing, and varies by a factor of $e^3 \approx 20$ from left endpoint to right. All derivatives of e^x equal e^x itself, so $\max |f^{(k)}| = e^3$ for every k , and the ratio of average to maximum is the same factor of about three for every derivative. Both Trap and Gauss converge cleanly, and the integrand is easy enough that the Gauss rule reaches machine precision with only $n = 512$ subintervals.

Integral (2), $\int_0^{\pi/3} e^{\sin(2x)} \cos(2x) dx$, is smooth with moderate oscillation and product structure. Its integrand contains two nonlinear factors—an exponential of a sinusoid, and a second sinusoidal factor—so the derivatives involve products that grow in combinatorial complexity but are bounded

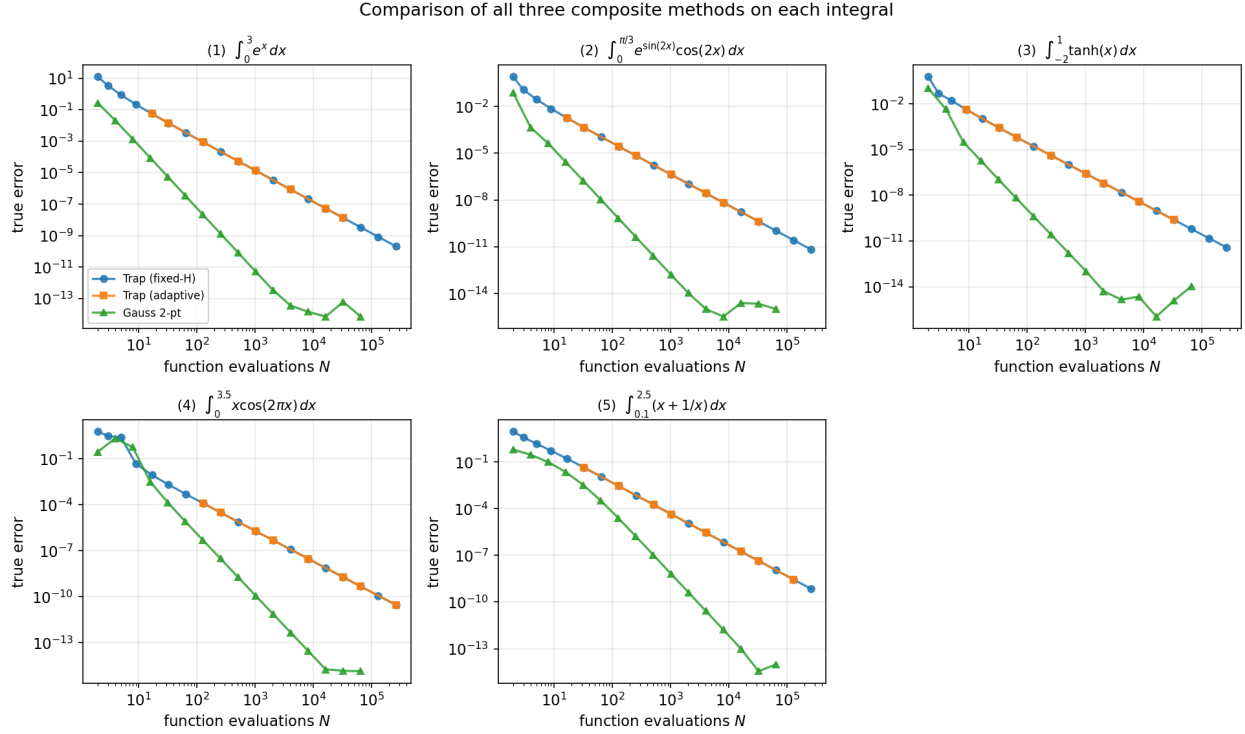


Figure 6: All three methods on common axes. The two Trapezoidal curves (fixed- H blue, adaptive orange) are nearly coincident on the $O(N^{-2})$ slope; the Gauss curve (green) follows the steeper $O(N^{-4})$ slope down to machine precision. The flat tail on the Gauss curves (integrals 1,2,3) marks round-off saturation.

by moderate constants. Because the interval is short ($\pi/3 \approx 1.05$) and the amplitude of the integrand is of order one, the total integral value $|I| \approx 0.69$ is small enough that even tight absolute tolerances correspond to reasonably achievable n values. Gauss achieves machine precision on this integrand by $n = 256$, slightly faster than on integrand (1) because the interval is shorter.

Integral (3), $\int_{-2}^1 \tanh(x) dx$, is smoothly monotone with saturating asymptotics; $|f''(x)| = |-2 \operatorname{sech}^2(x) \tanh(x)|$ peaks at a modest value of about 0.77 and is very small over most of the interval. This is the easiest integral for both methods by a factor of roughly $20\times$ in evaluation count compared to integrand (1), simply because the small $\max |f''|$ directly shrinks the Trap error bound. The Gauss rule is similarly advantaged by a correspondingly small $\max |f^{(4)}| \approx 4$, so it reaches machine precision at $n = 256$.

Integral (4), $\int_0^{3.5} x \cos(2\pi x) dx$, is oscillatory, with roughly seven full periods of the cosine within the interval. The oscillation is what makes this integrand challenging: every derivative picks up an additional factor of 2π in magnitude, so $\max |f''|$ is approximately 138 and $\max |f^{(4)}|$ is approximately 5400. Both bounds are large because derivatives amplify oscillations. At the same time, however, the observed true Trap error is roughly 250 times smaller than the bound, a gap much larger than for any of the other integrands. The reason is that oscillations in f translate into oscillatory error contributions from successive subintervals, which partially cancel once H is small enough to resolve the period. This cancellation is a genuine numerical phenomenon that the $\max |f''|$ bound does not capture, because the bound is a pointwise upper estimate whereas the true error is a signed integral average. Before H becomes small enough to resolve the period, the Gauss rule exhibits a pre-asymptotic plateau, visible at $n = 4$ in the data, which disappears once

the oscillation resolution condition is satisfied.

Integral (5), $\int_{0.1}^{2.5} (x + 1/x) dx$, is near-singular at the left endpoint due to the $1/x$ term. The derivatives $|f^{(k)}(x)|$ are proportional to $1/x^{k+1}$, so they blow up rapidly as $x \rightarrow 0$: $\max |f''| = 2/0.1^3 = 2000$, $\max |f^{(4)}| = 24/0.1^5 = 2.4 \times 10^6$, and $\max |f^{(6)}| = 720/0.1^7 = 7.2 \times 10^9$. These enormous constants directly inflate both the a priori bound and the rate at which the true error decreases with refinement. This is the hardest integral for both methods and requires approximately $10\times$ more Trap subintervals than integrand (1) at every tolerance. Nevertheless, because the singularity is integrable ($\int (1/x) dx = \ln x$) and because only the *tail* of the derivative near $x = 0.1$ is problematic (the rest of the interval is smooth), the Gauss rule is still efficient: it reaches an absolute error of 10^{-7} with only 256 subintervals. For this integrand the structural advantage of Gauss over Trap is the largest of any integral in the study, quantitatively: at absolute error 10^{-8} , Gauss uses 1024 evaluations to the Trap rule's 131,073, a $128\times$ speedup.

Taken together, these five integrands illustrate that method choice in numerical quadrature is not governed by a single “difficulty” number but by the interaction between the integrand's derivative behaviour and the error formula of each method. Smooth, slowly varying integrands are easy for both methods; oscillatory integrands benefit from the cancellation inherent in trapezoidal integration but challenge any method's asymptotic regime; and near-singular integrands inflate all derivative bounds uniformly but still reward the higher-order Gauss rule because the rate advantage compounds multiplicatively with refinement.

6 Conclusions

This assignment has demonstrated the implementation, validation, and comparison of three composite quadrature methods—fixed- H Composite Trapezoidal, adaptive step-halving Composite Trapezoidal, and Composite Two-Point Gauss–Legendre. The principal conclusions are as follows.

First, the a priori Trapezoidal sizing strategy is reliable but conservative. The predicted H_{pred} from inverting (2) always produces an n whose true error lies below the a priori bound, but the bound can overstate the error by as much as $242\times$ for integrals with localized second derivatives (integral (4)) and $48\times$ for near-singular integrands (integral (5)). For smoothly varying integrands (1)–(3) the bound is close to saturated, with ratios of 3–7. The conservatism has a simple interpretation: $\max |f''|$ is an *endpoint* bound, whereas the true Trap error scales with an *average* $|f''|$ that is typically smaller. In practice the a priori bound is a safe but potentially wasteful sizing tool; one should be prepared to use $10\times$ fewer subintervals than the bound suggests and check the error a posteriori.

Second, adaptive step-halving is robust and essentially cost-equivalent to a priori sizing. The step-halving implementation using the doubling identity (5) uses the minimum possible number of function evaluations ($2^L + 1$ after L refinements). The Richardson estimate $|T_{2n} - T_n|/3$ tracks the true error to within $\pm 1\%$ in all tested cases where refinement has progressed beyond the first two levels. In terms of accuracy-per-function-evaluation the adaptive and fixed- H schemes are essentially indistinguishable: their error-versus- N curves are nearly coincident on every integral. The adaptive scheme's decisive advantage is that it requires *no a priori knowledge of the integrand*: the user supplies a tolerance, and the scheme self-calibrates. This makes adaptive step-halving the preferred default for black-box integrands where derivative bounds are unavailable.

Third, the Composite Two-Point Gauss–Legendre Rule dominates both Trapezoidal variants for smooth integrands. Its $O(N^{-4})$ convergence, versus $O(N^{-2})$ for Trap, translates into a $16\times$ reduction of error per doubling of N , versus $4\times$ for Trap. The empirical efficiency advantage grows geometrically with the required accuracy: $2\text{--}8\times$ at tolerance 10^{-2} , $16\text{--}64\times$ at 10^{-6} , and over $100\times$ at 10^{-10} . Gauss reaches machine precision with $n \leq 1024$ for every integral tested, whereas the Trapezoidal variants cannot practically reach 10^{-10} accuracy for any integral within the tested budget of $n = 2^{18}$.

Fourth, the observed convergence rates match theory to high precision. Empirical rate analysis confirms $\hat{p} \rightarrow 2$ for Trap and $\hat{p} \rightarrow 4$ for Gauss, reaching the asymptote by $n \sim 64$ for well-conditioned integrands and by $n \sim 256$ for integrand (5) with large $\max |f^{(4)}|$. The convergence-rate plot (Figure 1) is a clean corroboration of the theoretical error expressions.

Fifth, integrand structure dominates method choice in the pre-asymptotic regime. For oscillatory integrand (4), all three methods show a small “pre-asymptotic” region at very small n where the rule cannot resolve the oscillations and the error does not decrease cleanly; once H is small enough to sample each period at least a few times, the asymptotic rates take over. Similarly, for near-singular integrand (5), the large pre-asymptotic constants delay the onset of asymptotic behaviour by an order of magnitude in n . These are not failures of the methods but reflections of the integrand’s intrinsic difficulty: asymptotic error analysis is only valid once H is small relative to the length scale of the integrand’s variations.

Sixth, round-off eventually limits every rule. The Gauss rule saturates near $10^{-14} \times |I|$ in every case tested; further refinement degrades the result slightly. This floor is intrinsic to the floating-point arithmetic and cannot be eliminated by method refinement; it would be slightly lower if compensated summation (Kahan summation) were used, but the gains would be modest. For practical accuracy requirements below 10^{-13} , extended precision arithmetic is required regardless of method.

Seventh, basic-rule quality dominates adaptive cleverness. The most important lesson of this study is that the Gauss rule with straightforward uniform subintervals beats the adaptive Trapezoidal rule by orders of magnitude on every integrand tested. Adaptive step refinement is not a substitute for using a higher-order basic rule; it is an orthogonal improvement that delivers its maximum benefit when layered on top of a high-accuracy basic rule. A natural extension of this work would be to implement an adaptive-step 2-point Gauss rule, which should combine the $O(H^4)$ accuracy of Gauss with the self-calibration of the adaptive scheme. Alternatively, a higher-order Gauss rule (3-point, 4-point) would further accelerate convergence, at the cost of making the derivative bound in the a priori error expression depend on even higher derivatives of f .

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